



Oral Disease Classifier

An end-to-end deep learning project: from a messy Kaggle dataset to a live AI web app anyone can try

TensorFlow / Keras

4 architectures compared

Deployed on Streamlit

Try it live → oral-disease-detection-cnn.streamlit.app

WHY THIS MATTERS

Oral disease is common, and early screening isn't always accessible

~45%

of people worldwide are affected by some form of oral disease (WHO)

6

common conditions this model screens for from a single photo

1 photo

is all the deployed app needs to return a prediction in seconds

Idea: package a trained classifier behind a simple upload-a-photo interface, so anyone can try an early, informal screening in their browser.

From a messy raw dataset to clean, leak-free training data

1

Cleanse

Dropped pre-augmented copies & YOLO-annotation folders, keeping only raw source images per class

2

Restructure

Consolidated into one clean folder per class (6 classes total)

3

Split 80/20

Randomized, leaked-proof train/val split (seed=42) via split-folders

4

Balance

`class_weight="balanced"` during training — validation set stays untouched & representative

5

Augment

Flip, rotation, zoom, contrast, brightness, noise — applied only to the training set

4 architectures trained and benchmarked head-to-head

Custom CNN

Built from scratch — 4 conv blocks (32→256 filters), BatchNorm, increasing dropout, GAP head. Tuned via a learning-rate × dropout grid search.

EfficientNetB0

ImageNet transfer learning, frozen base + custom classification head

ResNet50

ImageNet transfer learning, frozen base + custom classification head

MobileNetV2

ImageNet transfer learning, frozen base + custom classification head

Trained with Adam + sparse categorical crossentropy, class weights, and ReduceLROnPlateau / EarlyStopping / ModelCheckpoint callbacks.

RESULTS

Evaluated on F1-macro, not just accuracy, because classes are imbalanced

Model	Val Accuracy	Val Loss	F1 (macro)
 ResNet50	78.1%	0.571	77.6%
Auto-selected as the production model based on F1-macro across all 4 candidates.			

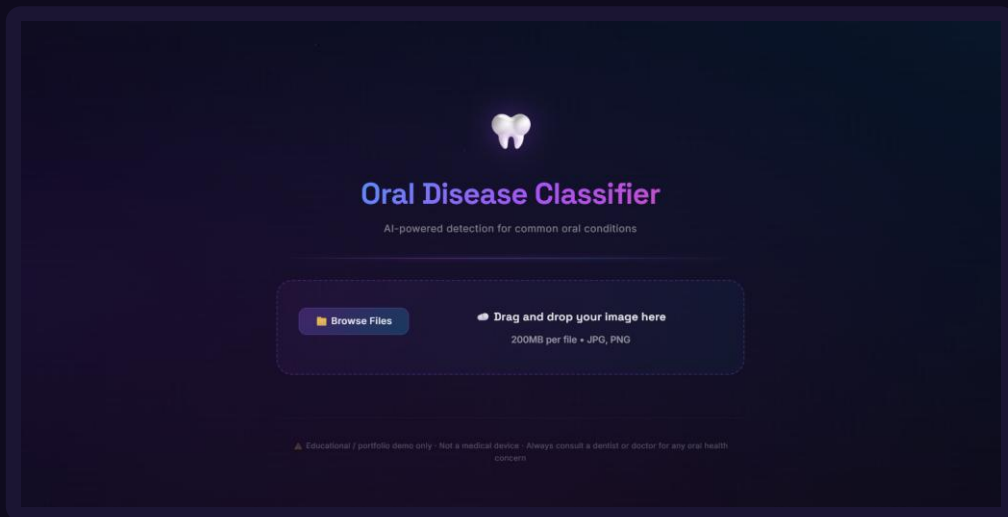
Exported as `model.keras` + `metadata.json` (class names, image size, preprocessing config) for a clean, reproducible inference step.

Why F1-macro?

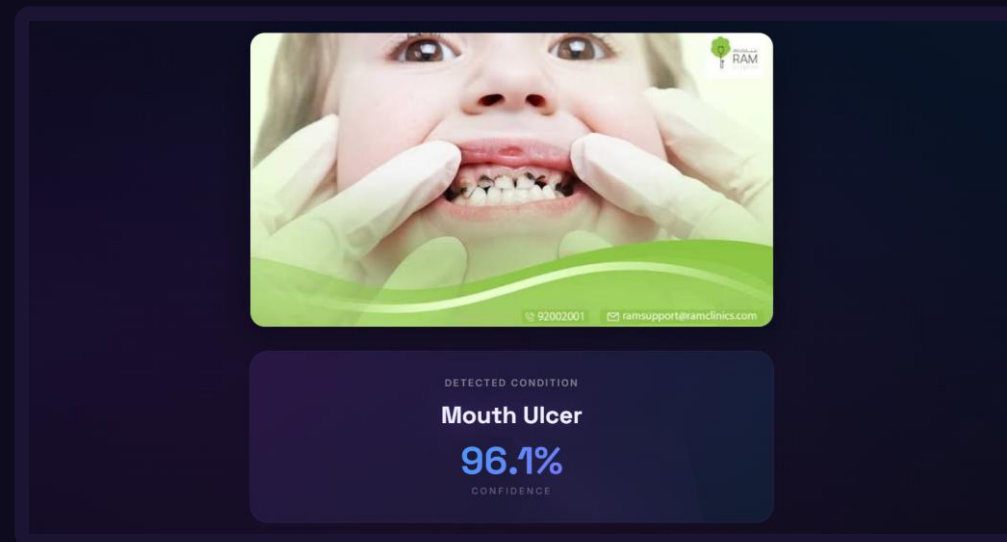
Minority classes like Caries and Mouth Ulcer have far fewer images than Gingivitis. Accuracy alone can hide poor performance on them — macro-F1 weights every class equally.

LIVE DEMO

From upload to prediction in seconds



1. Upload a photo



2. Get the detected condition + confidence

Full probability breakdown, not just a single label



Tech stack

- TensorFlow / Keras
- scikit-learn (metrics)
- NumPy · Pandas
- Streamlit
- Streamlit Community Cloud

All 6 class probabilities are shown, ranked — so the result is transparent, not a black-box label.

Try it yourself

Upload a photo, get an instant prediction across 6 oral conditions



Live demo

oral-disease-detection-cnn.streamlit.app



Source code

github.com/ahmed-rdwan/oral-disease-detection

⚠ Educational / portfolio demo only — not a medical device. Always consult a dentist or doctor for real oral health concerns.